

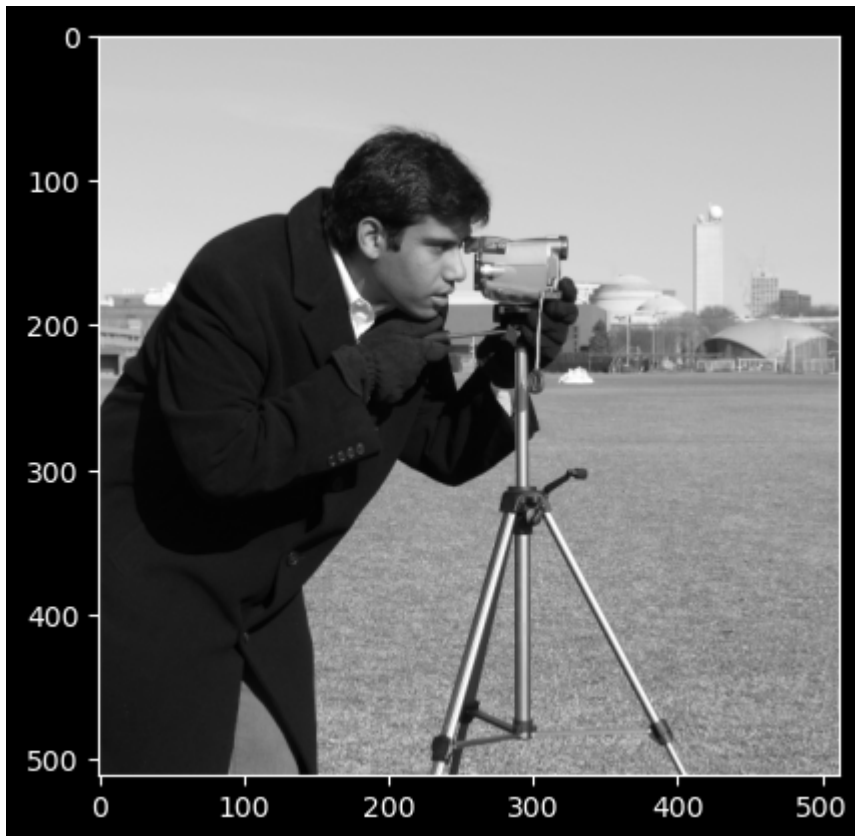
Lab 1

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```
In [1]: import cv2  
img = cv2.imread('images/cameraman.tif', cv2.IMREAD_GRAYSCALE) #Load the
```

```
In [2]: import cv2  
import matplotlib.pyplot as plt  
  
plt.imshow(img, cmap='gray')  
plt.show()
```



```
In [3]: from PIL import Image  
import matplotlib.cm as cm  
  
img2 = Image.open('images/lena_gray_256.tif')  
plt.imshow(img2, cmap=cm.Greys_r)  
plt.show()
```



```
In [4]: cv2.imwrite('output/new_image.jpg', img);
```

```
In [5]: img2.save('output/new_image2.jpg')
```

```
In [6]: print(img.shape)
```

```
(512, 512)
```

```
In [7]: print(img)
```

```
[[200 200 200 ... 189 190 190]
 [200 199 199 ... 190 190 190]
 [199 199 199 ... 190 190 190]
 ...
 [ 25  25  27 ... 139 122 147]
 [ 25  25  26 ... 158 141 168]
 [ 25  25  27 ... 151 152 149]]
```

```
In [8]: import numpy as np
```

```
img2_array = np.array(img2)
print(img2_array.shape)
```

```
(256, 256)
```

```
In [9]: print(img2_array)
```

```
[[162 162 162 ... 168 171 155]
 [162 162 162 ... 168 171 155]
 [162 162 162 ... 168 171 155]
 ...
 [ 48  57  54 ...  75  97  88]
 [ 46  50  52 ...  89 100  98]
 [ 44  55  54 ...  97 100 105]]
```

In []: